Detection and Estimation of Daily Oceanic Mesoscale Eddies From Spaceborne Global Navigation Satellite System-Reflectometry

Jin Xing, *Graduate Student Member, IEEE*, Feng Wang, Dongkai Yang, Chuanrui Tan, Xiangchao Ma, and Wenqian Chen

***Abstract*— The potential of spaceborne Global Navigation Satellite System-Reflectometry (GNSS-R) is explored for detecting and estimating global mesoscale oceanic eddies. A fractional Fourier transform-residual network (F-ResNet) model is used to achieve centimeter-level absolute dynamic topography (ADT), demonstrating a root mean square error (RMSE) of 5.05 cm, which is subsequently used for detecting oceanic eddies based on their height characteristics. Using CYGNSS data from 2020, comprising over 750 million samples, the research demonstrates that GNSS-R technology effectively supports daily oceanic eddy detection, achieving precision rates of 50.63% and 47.80%, recall rates of 89.95% and 90.48%, $F1$ scores of 64.79% and 62.56%, and accuracy rates of 47.92% and 45.52% for cyclonic and anticyclonic eddies, respectively. Although the method proves both effective and feasible, its performance remains suboptimal. To improve detection accuracy, the study explores high-bandwidth signals, whose sharper autocorrelation function enhances ADT retrieval and improves oceanic eddy detection accuracy. Simulation results from the ACE-BOC (15, 10) signal suggest precision rates of 85.51% and 84.14%, recall rates of 91.93% and 91.11%, $F1$ scores of 88.76% and 87.48%, and accuracy rates 79.80% and 77.75% for cyclonic and anticyclonic eddies, respectively. Moreover, supplementary metrics for eddy features indicate significant improvements with high-bandwidth signals. The application of these two signals for sustained, long-term monitoring of oceanic eddies in a specific region is also demonstrated, providing a foundation for eddy tracking. The tracking results show that the ACE-BOC (15, 10) signal can effectively track eddy trajectories, achieving an RMSE of 1.11 km, whereas CYGNSS yields an RMSE of 8.4 km. Finally, this article discusses the characteristics of ACE-BOC and carrier-altimeter signals, highlighting the potential of advanced spaceborne GNSS-R altimeters and discussing future directions in this field.**

***Index Terms*— ACE-BOC (15, 10) signal, carrier-altimetry, code-altimetry, Global Navigation Satellite System-Reflectometry (GNSS-R), mesoscale eddies.**

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I. Introduction

NUMEROUS eddies exist in the world's oceans, characterized as currents with circular motions around specific centers, and diameters ranging from several hundred meters to hundreds of kilometers. The vertical convection and horizontal motion of these eddies are crucial in the large-scale transport of mass, heat, and biogeochemical materials, both regionally and globally. Consequently, eddies—particularly mesoscale eddies with diameters ranging from 10 to 100 km—are essential for the distribution of oceanic resources [1]. The global identification of eddies has become a critical topic in oceanography [2]. Eddies induce anomalies in sea surface height (SSH), sea surface temperature (SST), and material transport; thus, detecting and tracking these anomalies are essential for identifying eddies. Satellite altimetry, capable of measuring global SSH, has proven effective in detecting and monitoring mesoscale eddies on a global scale [3]. Due to the limited temporal resolution of a single satellite altimeter, it is essential to combine global SSH data from multiple satellites to derive daily mean-gridded sea level anomaly (SLA) data. As a typical active radar system, satellite altimetry requires a high-cost, high-power transmitter, making the concurrent development of a satellite altimeter constellation financially challenging.

Several potential candidates must be investigated to improve satellite altimeter data. A novel ocean altimeter concept has been proposed that uses signals from the Global Navigation Satellite System (GNSS) [4]. This technology, known as GNSS-Reflectometry (GNSS-R), uses reflected GNSS signals from the sea surface as a radar signal source to measure or retrieve various oceanic physical parameters, including sea state [5], [6], SSH [7], [8], and sea ice [9], [10]. Without the need for an independent transmitter, this system offers lower costs and reduced power consumption compared with traditional active microwave remote sensing. It can be integrated into various cost-effective, power-constrained platforms, enabling the development of a networked observation system. With the development of various spaceborne missions, including UK-TDS [11], CYGNSS [12], BuFeng-1 A/B [13],

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FengYun-3E [14], and the Spire constellation [15], GNSS-R has been used to provide global measurements of Earth's physical parameters. A detailed analysis of the ocean altimetry precision of spaceborne GNSS-R was presented in [8]. The results indicated that the GPS C/A signal provides meter-level precision, while the L5 signal and L1 full signal offer decimeter-level precision. The first spaceborne SSH measurement using GNSS-R was derived from UK TDS-1 data [16], achieving root mean square errors (RMSEs) of 8.1 and 7.4 m when comparing TDS-1 SSH with the DTU10 mean SSH (MSSH) for South Atlantic and North Pacific, respectively. The GPS C/A signal altimetry from CYGNSS achieved a precision of 4.3 m and an accuracy of 1.9 m. In comparison, the Galileo E1 and BeiDou-3 B1 signals exhibited superior altimetry performance, with an improvement factor of approximately 1.8 over the GPS C/A signal [17]. Due to its configuration, GNSS-R must account for additional factors compared with active altimetry to deliver more accurate measurements. Given the robust capabilities of multivariate nonlinear regression, various machine learning techniques have been applied to assess SSH in spaceborne GNSS-R. A fractional Fourier transform-residual network (F-ResNet) model was trained to achieve higher precision in GNSS-R altimetry compared with the geometry-based approach [18], with an RMSE of 1.17 m achieved for CYGNSS data. In addition, the carrier phase provides more precise altimetry than the pseudorandom noise (PRN) code. A spaceborne GNSS-R carrier-altimeter has been proposed and demonstrated. In [19] and [20], the altimetry performance of the reflected GNSS signal carrier from sea ice and lake surfaces was evaluated, revealing centimeter-level precision. Similarly, centimeter-level precision with a 1-Hz update rate has been achieved for SSH [21]. The carrier-altimeter relies on signal coherence, and the results in [21] indicated that under grazing angle (90° incident angle) geometries and with wind speeds below 6 m/s, only 33% of the tracks demonstrated coherence. In most cases, the spaceborne GNSS-R carrier-altimeter cannot be widely used for SSH because of the strict conditions. Nonetheless, current research on spaceborne GNSS-R for SSH measurement illustrates its capability to deliver worldwide SSH data.

A significant amount of study has concentrated on quantifying SSH with spaceborne GNSS-R data. Nevertheless, limited research has investigated the identification, estimation, and monitoring of oceanic eddies using SSH data retrieved from spaceborne GNSS-R. The preliminary eddy detection experiment conducted from a low-altitude aircraft achieved the RMSE of 10 cm with a spatial resolution of 20 km for eddy observation [22]. Similar observation capabilities from the airborne platform were reported in [23]. These studies highlight the potential contribution of airborne GNSS-R to oceanic eddy detection. The first spaceborne GNSS-R observations of mesoscale eddies were presented in [24] and [25], where the observable used was the normalized bistatic radar cross section (NBRCS) derived from CYGNSS data, rather than SSH. It was found that NBRCS exhibited a clear inverse correlation with surface heat flux and stress, both of which are influenced by eddies. Furthermore, the response of NBRCS from spaceborne GNSS-R to current signatures was reported in [26]. This study also demonstrated the potential of spaceborne GNSS-R carrier-altimeter systems for observing SLA mapping in loop current regions [27]. Although these investigations concentrated on individual eddy instances rather than a global perspective, they effectively recognized eddy signatures in spaceborne GNSS-R data rather than directly detecting and characterizing the eddies. The present work aims to detect and estimate global mesoscale eddies using spaceborne GNSS-R data.

The main contributions of this study are summarized as follows.

1) An F-ResNet is used to retrieve the ADT from spaceborne GNSS-R observables with a centimeter precision of 5.05 cm, achieving a 96.75% improvement over the traditional geometric method.
2) The potential of spaceborne GNSS-R for daily and global detection and estimation of oceanic eddies using CYGNSS data is first demonstrated, achieving precision rates of 50.63% and 47.80% and accuracy rates of 47.92% and 45.52% for cyclonic and anticyclonic eddies, respectively.
3) The capability for high-precision eddy detection is first demonstrated and assessed using ACE-BOC (15, 10) signals. Daily assessments indicate precision rates of 85.51% and 84.14% and accuracy rates of 79.80% and 77.75% for cyclonic and anticyclonic eddies, demonstrating a 70% improvement relative to the CYGNSS GPS L1 signal. The findings highlight the potential of high-bandwidth spaceborne GNSS-R signals, indicating an encouraging path for future development.

The following sections of this article are structured as follows. Section II describes the methodology for assessing absolute dynamic topography (ADT) using spaceborne GNSS-R data. Section III presents the methodologies for detecting and estimating mesoscale eddies. Section IV uses actual CYGNSS data to identify and quantify global eddies while assessing the effectiveness of detecting and estimating eddies through a next-generation wideband GNSS signal. Sections V and VI present the discussion and conclusion, respectively.

II. Measurement of ADT

A. Traditional Geometric Methods

1) Sea Surface Height: The along-trajectory SSH can be derived from the geometric relationships associated with each reflection event. The measurement method of SSH using spaceborne GNSS-R relies on the geometric configuration and the propagation delay between direct and reflected GNSS signals, which can be described as

$$\hat{h}_{\text{ssh}} = \frac{\Delta\tau_{\text{measured}} - \Delta\tau_{\text{modeled}}}{2 \cdot \cos\theta} \tag{1}$$

where θ is the incident angle of the GNSS signal at the specular point; $\Delta\tau_{\text{measured}}$ is the propagation delay measured from the arrival times of the direct and reflected GNSS signal; and $\Delta\tau_{\text{modeled}}$ is the propagation delay modeled based on the Earth's ellipsoid model. The arrival time of the direct signal is determined using a closed-loop delay-locked technique. However, due to the low signal-to-noise ratio (SNR) and the coherent and incoherent components, obtaining the arrival time

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of the reflected signal is challenging. The standard method for estimating the arrival time of the reflected signal involves tracking a specific point in the delay power waveform, such as the first derivative of the waveform (DER), or by fitting the power waveform to a theoretical model [17]. The position of a specific point in the delayed waveform is influenced by both SSH and the sea state, as well as the incident angle [28]. As a result, the bias and standard deviation of the estimated arrival time are time-dependent and geometry-dependent. The tracking accuracy of the best-fitting approach is closely related to the theoretical model, which is a function of the observation and scattering scenarios. Therefore, the best-fitting method requires an accurate theoretical model and precise auxiliary data. Compared with the single-point approach, the best-fitting method demonstrates superior tracking performance. The modeled propagation delay is derived from the Earth ellipsoid model, accounting for ionospheric and tropospheric delays as described in [28]

$$\Delta\tau_{\text{modeled}} = (R_{\text{ts}} + R_{\text{rs}} - R_{\text{tr}}) + \delta_{\text{iono}} + \delta_{\text{trop}} + \delta_{\text{orbit}} + \delta_{\text{ant}} \tag{2}$$

where R_{ts} and R_{rs} represent the distances from the GNSS and LEO satellites to the specular point, respectively; R_{tr} denotes the distance between the GNSS and LEO satellites; δ_{iono} and δ_{trop} correspond to the ionospheric and tropospheric delays, respectively [28], [29]; δ_{orbit} represents the error of the orbit; and δ_{ant} denotes the baseline bias between the two antennas.

2) Absolute Dynamic Topography: The basic observable to detect eddies is ADT which can be subtracted from the geoid height from the altimetric MSSH above the reference ellipsoid as

$$h_{\text{adt}} = \overline{h}_{\text{ssh}} - h_{\text{geoid}} \tag{3}$$

where $\overline{h}_{\text{ssh}}$ denotes the averaged SSH over a specific time period, typically one day; and h_{geoid} represents the geoid height. At a given observation time, SSH is the MSSH modulated by tidal effects, as follows:

$$h_{\text{ssh}} = \overline{h}_{\text{ssh}} + h_{\text{tide}} \tag{4}$$

where h_{tide} represents the tidal height. From (3) and (4), it follows that the ADT can be estimated from the measured SSH as

$$\begin{aligned} h_{\text{adt}} &= \left\langle \hat{h}_{\text{ssh}} - h_{\text{tide}} \right\rangle - h_{\text{geoid}} \\ &= \left\langle \hat{h}_{\text{ssh}} - h_{\text{tide}} - h_{\text{geoid}} \right\rangle \end{aligned} \tag{5}$$

where $\langle\cdot\rangle$ denotes the mean operator. The along-trajectory ADT can be derived by subtracting the geoid height and tidal effects from the measured SSH.

B. F-ResNet Method

As demonstrated in Section II-A, SSH and ADT are related through a fixed geometric relationship. SSH represents the height of the sea surface relative to a reference ellipsoid, covering a wide range of values (from −100 to +80 m). In contrast, ADT is derived from SSH data, combined with oceanographic models and other auxiliary information, typically ranging from

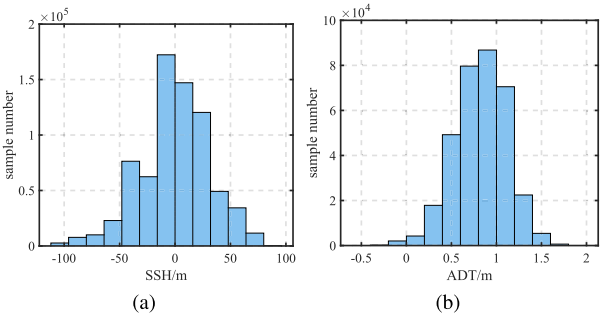


Fig. 1. Samples distributions of (a) SSH and (b) ADT within their respective ranges in a global daily map.

−0.2 to +1.7 m, as shown in Fig. 1. Although these two quantities are geometrically related, their error characteristics and precision vary significantly. The objective of using neural networks for SSH prediction is to minimize the prediction error associated with SSH. The extensive variability of SSH results in prediction errors generally measured in meters. Converting SSH predictions into ADT involves the propagation of associated errors through a fixed geometric relationship, typically yielding precision at the meter scale. When the neural network is trained to predict ADT directly, it optimizes its predictions for ADT errors, thereby modifying the model's learning strategy and enhancing accuracy.

The smaller range of ADT, compared with SSH, allows the neural network to better capture subtle local variations, with the ADT field exhibiting more distinct, nuanced changes. This enables the network to effectively learn fine-scale variations, achieving higher prediction accuracy, typically within the decimeter range. Even with identical input data and a consistent network architecture, transitioning from SSH to ADT as the target variable enhances the model's precision, especially in detecting fine-scale variations. This underscores the importance of directly predicting ADT rather than relying on SSH-derived estimates to improve the accuracy of predictions, particularly in capturing complex oceanic dynamics.

The residual bias of the ADT is established as time-dependent and geometry-dependent, influenced by sea state and incident angle. The delayed waveform, or delay–Doppler map (DDM), contains information regarding the sea state in addition to the propagation delay. The final measured ADT is presented as follows:

$$h_{\text{adt}} = \hat{h}_{\text{adt}} + b(\text{DDM}, \theta) \tag{6}$$

where DDM denotes the DDM of spaceborne GNSS-R. Developing an explicit theoretical or empirical model for the residual bias is challenging. Machine learning can be used to regress (6). In this article, the machine learning approach used is the F-ResNet, as proposed in [18]. In this network, the DDMs undergo a fractional Fourier transform (FrFT) to efficiently distinguish between the signal and noise components. Following the application of Gaussian filtering, the DDMs are subjected to an inverse FrFT. The power ratio (PR) is derived from DDM. In addition, auxiliary data, including PR, incidence angle, SNR, vector velocity and location of the GPS satellite, vector velocity and location of the CYGNSS

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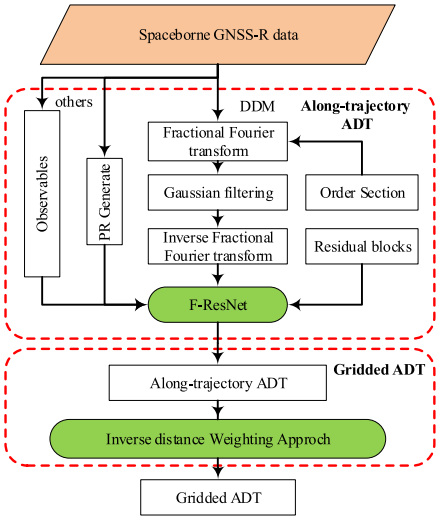


Fig. 2. Illustration of obtaining gridded daily mean ADT from spaceborne GNSS-R.

satellite, geoid height at the estimated specular point, and SSH computed by the traditional method, are gathered. The processed features are subsequently input into a developed network incorporating residual blocks for enhanced learning. The F-ResNet produces the along-trajectory ADT as its final output. The upper half of Fig. 2 illustrates that this multi-step process successfully integrates signal separation, feature extraction, and deep learning techniques to achieve precise ADT estimation.

C. Gridded Daily Mean ADT

Gridded ADT refers to the process of mapping irregularly distributed along-trajectory ADTs onto a regular spatial grid, thereby producing standardized and structured ADT products. Given the spatial resolution of the GPS L1 CA signal, the Earth's surface in this study is gridded at a resolution of $0.25° \times 0.25°$. The ADT for each grid cell is computed as the daily mean of the along-trajectory ADTs for specular points that fall within the grid. Due to the nonuniform distribution of specular points over the sea surface, certain grid cells may lack sufficient ADT samples. In these instances, the ADT values for these grid cells are derived by 2-D interpolation. In this study, the inverse distance weighting method is used to address such gaps

$$z(x) = \sum_{i=1}^{n} w_i z(x_i) \tag{7}$$

where $z(x_i)$ is the averaged ADT in the neighborhood of the gap at x; and w_i is the corresponding weight of the grid point x_i, calculated as

$$w_i = \frac{\frac{1}{d^p(x,x_i)}}{\sum_{i=1}^{n} \frac{1}{d^p(x,x_i)}} \tag{8}$$

where $d(x, x_i)$ is the distance between x and x_i; and p is a natural number. The detailed procedure for computing the gridded daily mean ADT is illustrated in Fig. 2.

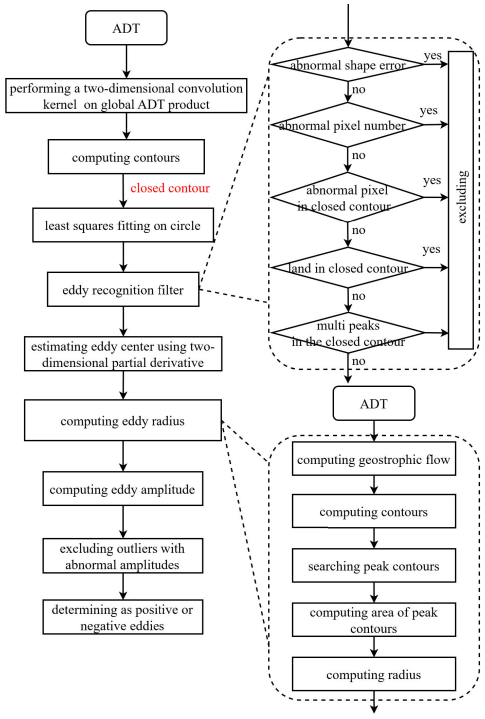


Fig. 3. Procedure for detecting and estimating eddy.

III. Methodologies of Detecting and Estimating Eddies

A. Detecting and Estimating Approaches

Eddies can be detected from the ADT product, and several algorithms for eddy detection have been developed. The method used in this study is based on the approach proposed by [30], which is based on the dynamical field and closed contours of a threshold. The detailed procedure for eddy detection is given in Fig. 3. The dynamical field used here is the global ADT product. Before detecting eddies, a 2-D Bessel convolution kernel is used to smooth the ADT. The contours are subsequently generated from the smoothed ADT at 1-cm intervals, and each contour is crossed in succession. Five criteria, derived from the shape and internal attributes of the closed contour, are used to determine whether the closed contour qualifies as an eddy, as detailed below [31].

1) The shape error of the closed contour, which is defined as follows [32], should be below a given threshold

$$E_{\text{shape}} = \frac{A_c + A_{\text{effc}} - 2 \cdot A_{\text{inter}}}{A_{\text{effc}}} \times 100\% \tag{9}$$

where A_c and A_{effc} are, respectively, the areas of the closed contour and fit circle.

2) The pixel number in the closed contour should be limited to a proper range.
3) Only one local ADT minimum/maximum is contained in the closed contour for cyclonic/anticyclonic eddies.
4) The closed contour should not contain a land or island.
5) The ADTs in the closed contour should be below/above the current ADT interval for cyclonic/anticyclonic eddies.

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In addition to shape errors, the characteristics of the eddy are described using its center, radius, and amplitude. The center of the eddy is defined as the position of either the minimum or maximum within the closed contour. The radius is represented by the equivalent radius corresponding to the radius of the best-fitting circle that encompasses the maximum geostrophic flow. The geostrophic flow is derived from the ADT as

$$V_\theta = \sqrt{U^2 + V^2} \tag{10}$$

where U and V are the geostrophic flow components along the east and north directions, which are computed through performing partial differentiation on the ADT as

$$\begin{cases} U = -\dfrac{g}{f}\dfrac{\partial(\mathrm{ADT})}{\partial n} \\ V = \dfrac{g}{f}\dfrac{\partial(\mathrm{ADT})}{\partial e} \end{cases} \tag{11}$$

where g is the gravitational acceleration; f is the Coriolis parameters; and e and n, respectively, represent the east and north directions. The amplitude of the eddy is the absolute value between the ADT at the eddy center and the averaged ADT along the closed contour as

$$A_{\mathrm{eddy}} = \left| h_{\mathrm{adt}}^{\mathrm{center}} - \left\langle h_{\mathrm{adt}}^{\mathrm{edge}} \right\rangle \right| \tag{12}$$

where $h_{\mathrm{adt}}^{\mathrm{center}}$ is the ADT at the center of the eddy; and h_{adt} is the ADT at the closed contour.

B. Assessment Metrics

The detection of eddies is a typical classification problem. The following metrics are used to quantitatively assess the performance of eddy detection.

1) Precision

$$p_{\mathrm{pre}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \times 100\%. \tag{13}$$

True positive (TP) denotes the eddies that were correctly detected. False positive (FP) denotes the areas that were incorrectly detected as eddies (they do not actually have eddies). In this study, two eddies are considered successfully matched, categorized as TP, if the GNSS-R-detected results overlap with the true eddy locations from the Archiving, Validation and Interpretation of Satellite Oceanographic (AVISO) reference dataset, as shown in Fig. 4. Samples are supposed to be eddies to demonstrate different scenarios. The similarity coefficient (SC) decreases as the distance between the two eddies increases, ultimately reaching 0, which signifies a lack of correspondence and implies that the eddy is not matched by the methodology used in this article. The metric measures the proportion of correctly detected eddies out of all the areas marked as eddies by the model.

2) Recall

$$p_{\mathrm{rec}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \times 100\%. \tag{14}$$

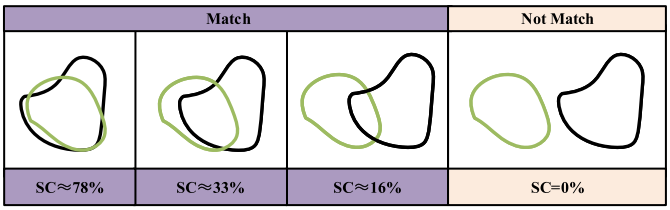


Fig. 4. Schematic of the detected eddies and the true eddies matching principle. (The green contour represents the actual eddy contour, whereas the black contour denotes the detected eddy contour.)

False negative (FN) denotes the eddies that were missed (they exist but were not detected). The metric measures the proportion of actual eddies that the model correctly detects.

3) $F1$ score

$$p_{F1} = 2\frac{p_{\mathrm{pre}} \cdot p_{\mathrm{rec}}}{p_{\mathrm{pre}} + p_{\mathrm{rec}}} \times 100\%. \tag{15}$$

The metric measures the harmonic mean of precision and recall, balancing FP and FN.

4) Accuracy

$$p_{\mathrm{acc}} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \times 100\%. \tag{16}$$

True negative (TN): Areas correctly identified as not containing eddies. In this method, TN = 0. Because it is not necessary for the model to identify noneddy regions as "correctly identified as noneddy." The metric measures the overall proportion of correct classifications (both eddies and noneddies) by the model.

To evaluate the characteristics of the detected eddies, they are compared with those from other data sources, which are considered ground truths. Five assessment metrics for comparing the two types of eddies are defined as follows.

1) Distance error (DE) between two eddy centers

$$E_D = \|\mathbf{P}_{\mathrm{detected}} - \mathbf{P}_{\mathrm{true}}\| \tag{17}$$

where $\mathbf{P}_{\mathrm{detected}}$ and $\mathbf{P}_{\mathrm{true}}$ are, respectively, the center positions of the detected and true eddies.

2) Radius error (RE)

$$E_R = r_{\mathrm{detected}} - r_{\mathrm{true}} \tag{18}$$

where r_{detected} and r_{true} are, respectively, the radiuses of the detected and true eddies.

3) Amplitude error (AE)

$$E_A = A_{\mathrm{detected}} - A_{\mathrm{true}} \tag{19}$$

where A_{detected} and A_{true} are, respectively, the amplitudes of the detected and true eddies.

4) Integrated relative error (IRE)

$$E_I = \sqrt{\left(\frac{E_A}{A_{\mathrm{detected}}}\right)^2 + \left(\frac{E_R}{r_{\mathrm{detected}}}\right)^2 + \left(\frac{E_D}{d_{\mathrm{ref}}}\right)^2} \tag{20}$$

where d_{ref} is a relative distance, which is set to 125 in this study based on empirical data.

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5) SC [31]

$$S = \frac{A_c \cap A_{\text{true}}}{A_c \cup A_{\text{true}}} \times 100\% \tag{21}$$

where A_{true} is the area of the true eddy; and $\cap$ and $\cup$ are, respectively, the intersection and union operators.

IV. Demonstration and Assessment

A. CYGNSS Measured Data

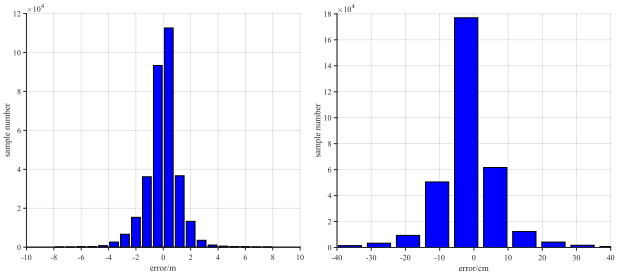
1) Data Sources: The CYGNSS data are used to detect and estimate global eddies. The oceanic products from AVISO are used to compare to the data source.

a) CYGNSS data: The CYGNSS aims to use reflected GNSS signals to monitor tropical cyclones, using an eight-satellite constellation that was launched into orbit in December 2016. The primary observable is the DDM of the reflected GPS L1 CA signal, sampled at 1 Hz. The data used for the code-altimeter are sourced from the CYGNSS L1-level Science Data Record (SDR) Version 3.1 dataset in 2020 (https://cmr.earthdata.nasa.gov/virtual-directory/collections/C2146321631-POCLOUD). In addition, the CYGNSS instruments are configured to acquire raw intermediate frequency (IF) signal samples from both direct and reflected signals. These raw data have been postprocessed by a software receiver and have been used to explore novel applications [21], [33], including the carrier-altimeter. The complex correlation value of these raw IF data, which can be downloaded from https://aliga.ice.csic.es/gold_rtr_mining/, is used for carrier-altimeter analysis.

b) AVISO data: The AVISO mission aims to develop and release radar altimeter data products jointly using AltiKa, CryoSat-2, Jason-2, Jason-1, and HY-2A altimeters. In this study, the MSSL and Eddy data (https://www.aviso.altimetry.fr/en/data.html) are used as the ground truth for comparison.

c) Other auxiliary data: The tidal data used in this research are derived from the global ocean tide model developed by Danmarks Tekniske Universitet (DTU), featuring a granular resolution of $0.125° \times 0.125°$ and incorporates the 12 principal tidal constituents. For the ionospheric delay model, the total electron content (TEC) data are sourced from the 2012 edition of the International Reference Ionosphere (IRI-2012) model, as documented in the literature [34], following the methodology outlined by Montenbruck and Gill [35]. The geoidal model used in this study is based on the Earth Gravitational Model of 1996 (EGM96). To mitigate the errors associated with GNSS orbital calculations, data from the International GNSS Service (IGS) are used to refine the GPS satellite positional data present in the SDR V3.1 dataset.

2) Results Analysis: First, the machine learning parameters were trained to retrieve along-trajectory ADTs. The CYGNSS data are uniformly distributed across 2020, using Jason-3 data as the reference label. Following temporal and spatial matching, a total of 164 737 samples are allocated into training, validation, and test sets in a ratio of 7:1.5:1.5 to develop the model. The RMSE of F-ResNet in measuring along-trajectory ADT is 12 cm, which represents a significant improvement over the traditional geometry-based method, which has an RMSE of 6.26 m. This improvement highlights the superior capability of F-ResNet in ADT measurement, as it integrates a more comprehensive set of information about sea surfaces from the DDM.



(a) (b)

Fig. 5. Distributions of errors of gridded ADT using (a) traditional geometric methods and (b) F-ResNet method.

Once the model is successfully trained, a large dataset of approximately 750 million records is used to retrieve the entire year along-trajectory ADTs, which are then used for the detection of eddies throughout the year. As for gridded ADT result, Fig. 5(a) illustrates the error distribution of the traditional geometric method at the meter level, with an RMSE of 1.54 m. Fig. 5(b) illustrates the error distribution of the smoothed ADT using F-ResNet. The predominant inaccuracies range from -20 to $+20$ cm, with an RMSE of 5.05 cm, reflecting a substantial enhancement of around 96.75%. Global maps illustrating the ground truth and the two methods for retrieving ADT on July 8, 2020 are illustrated in Fig. 6, demonstrating spatial consistency. The findings indicate that F-ResNet-derived ADT is extremely proficient in identifying large-amplitude oceanic eddies, whereas smaller amplitude eddies may provide greater difficulties in detection.

Fig. 7 presents an example of the detection of a large-scale cyclonic eddy with an amplitude of approximately 1.0 m. The closed contours of the ground truth exhibit an elliptical shape; however, those in the CYGNSS-based ADT show several irregularities, including bumps and depressions. These inconsistencies, which hinder accurate eddy detection, are primarily attributed to the limitations in GPS CA signal measurement. To specify the eddies in Fig. 7, the distance between the centers of the two eddies is 2 km, the error in eddy radius is 7.8 km, the AE is 7 cm, the IRE is 2.1, and the SC is 56%. These discrepancies between the two spatial ADT maps indicate that while CYGNSS-based ADT can effectively detect certain large-scale eddies, the shape characteristics of the detected eddies are subject to significant errors.

For the entire year of 2020, data were selected to detect global eddies and assess overall performance. The analysis focuses on the region between latitudes of $-40°$ and $40°$, which is fully covered by CYGNSS data. As illustrated in Fig. 8, a daily example, the detected eddies are generally consistent with the ground truth in terms of geographic location. However, significant discrepancies are observed in the shapes of the detected eddies when compared with the ground truth. Throughout the year 2020, a total of 870 688 cyclonic eddies and 838 913 anticyclonic eddies were identified in the

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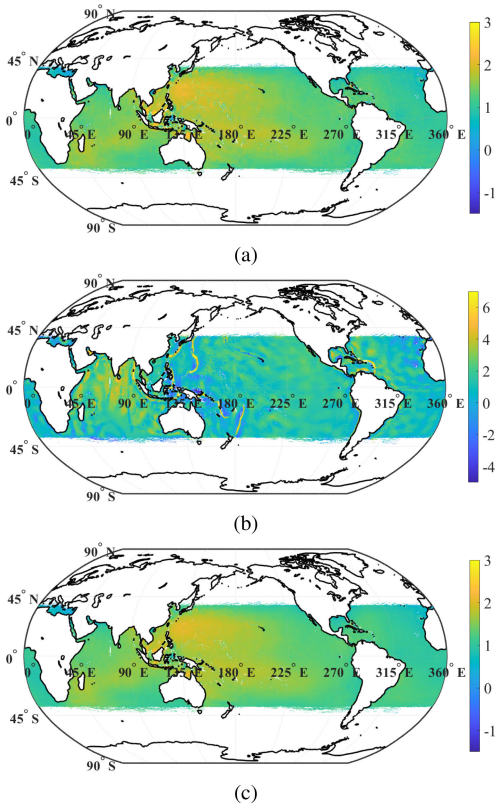


Fig. 6. Global maps of (a) ground truth, (b) traditional geometric method, and (c) F-ResNet method of ADT using CYGNSS on July 8, 2020.

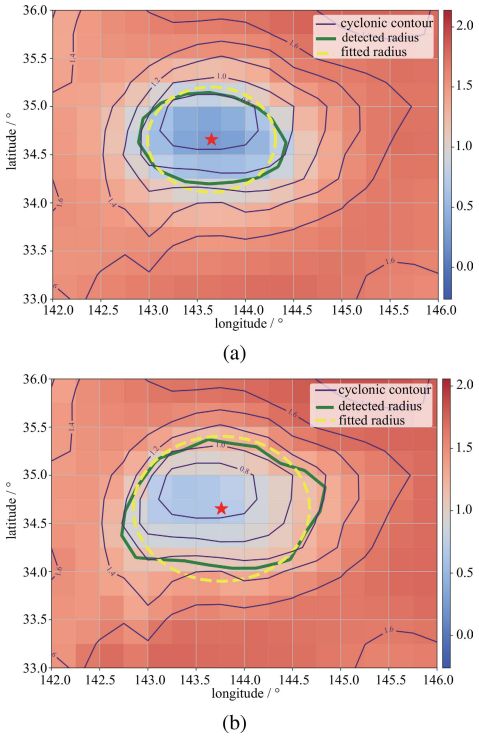


Fig. 7. Daily example on July 8 of the spatial distribution of (a) ground truth and (b) CYGNSS-based ADT.

reference dataset. The CYGNSS-based detection yielded 1 546 837 cyclonic eddies and 1 587 849 anticyclonic eddies. Among these, the true detection numbers were 783 204

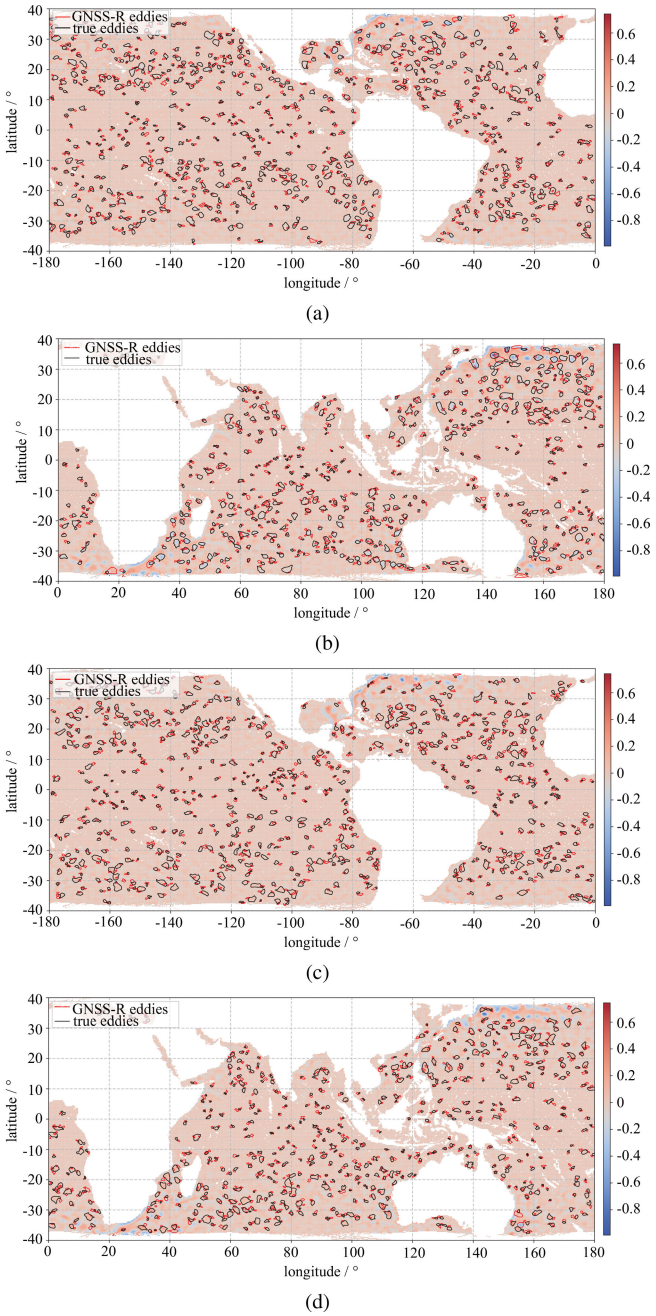


Fig. 8. Detected (a) and (b) cyclonic eddies and (c) and (d) anticyclonic eddies for (a) and (c) western hemisphere and (b) and (d) eastern hemisphere.

cyclonic eddies and 759 071 anticyclonic eddies. This result suggests that a significant proportion of the detected eddies were incorrectly identified. The corresponding performance metrics are summarized in Table I. The precision of the detected eddies is approximately 50%.

The shape error, defined in (9), quantifies the degree of similarity between an eddy and a standard circle. A lower shape error indicates that the eddy is more circular. Fig. 9 presents the distributions of shape errors for both CYGNSS-based eddies and ground-truth data. It is evident that most of the ground truths and CYGNSS-based eddies exhibit elliptical shapes. The shape errors for both the datasets primarily range from 25% to 45%. However, the distributions are notably

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TABLE I
DETECTION PERFORMANCE METRICS OF THE DETECTED CYCLONIC AND ANTICYCLONIC EDDIES

	cyclonic / %	anticyclonic / %
precision	50.63	47.80
recall	89.95	90.48
F1 score	64.79	62.56
accuracy	47.92	45.52

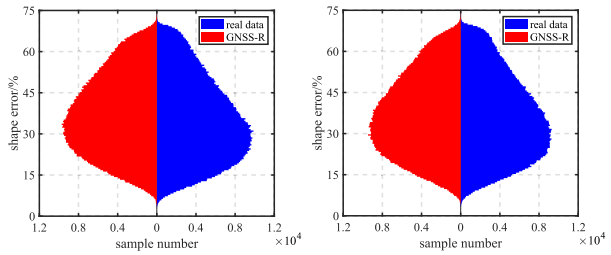


Fig. 9. Distributions of the shape errors for the CYGNSS-based (a) cyclonic and (b) anticyclonic eddies, as well as the ground truths.

distinct. The peak of the shape error distribution for the ground truths occurs at approximately 20%, while the peak for the CYGNSS-based eddies is closer to 30%. This discrepancy suggests that the CYGNSS-based eddies exhibit substantial shape differences compared with the ground truth. This conclusion is further supported by Fig. 10(a), which shows the distribution of differences between the shape errors of the ground truths and the CYGNSS-based eddies. The majority of these differences exceed 10%.

Fig. 10(b) presents the SCs between the detected eddies and the ground truth. The distribution is rather uniform, with no distinct declining tendency. The values ranging from 0.8 to 1 are marginally elevated compared with those approximately at 0.4. Nonetheless, most samples are concentrated below 0.15, indicating that the shapes of the detected eddies have very low similarity to the ground truths.

To further evaluate the shape differences between the two types of eddies, the center DEs, REs, and AEs between the ground truth and CYGNSS-based eddies are computed. Fig. 10(c) presents the distributions of the center DEs. The center DEs for both cyclonic and anticyclonic eddies display analogous distributions, with most errors being less than 50 km. As shown in Fig. 10(d), the AEs between the CYGNSS-based eddies and ground truths primarily fall within the range of -3 to 3 cm. Furthermore, both the distributions show positive means, indicating that the CYGNSS-based eddies are generally stronger than the ground truths. Fig. 10(e) illustrates the distributions of the REs. Similar distributions are observed for both cyclonic and anticyclonic eddies, with REs ranging from -50 to 200 km and showing a positive bias of approximately 25 km. These positive biases suggest that the shapes of the CYGNSS-based eddies are generally larger than those of the ground truths.

Fig. 10(f) shows the distributions of the IREs between the ground truth and CYGNSS-based eddies. The IREs

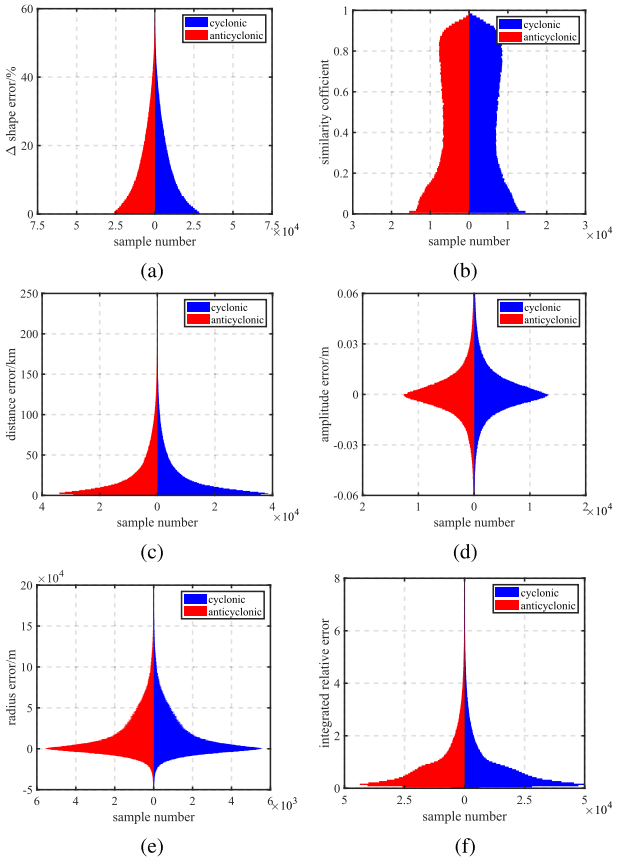


Fig. 10. Distributions of (a) shape error, (b) SCs, (c) center DEs, (d) AEs, (e) REs, and (f) IREs of the detected cyclonic and anticyclonic eddies.

for both cyclonic and anticyclonic eddies primarily range from 1 to 3. These performance metrics indicate that while the CYGNSS altimeter can detect some large-scale eddies, its detection performance is insufficient, with the shapes of the detected eddies significantly deviating from those of the true eddies. The spaceborne GNSS-R system, as a novel altimeter using narrowband signals like the GPS L1 CA signal, encounters difficulties in generating precise oceanic altimetry results. Consequently, the advancement and verification of next-generation spaceborne GNSS-R altimeters are essential for enhancing detection capabilities.

3) Eddy Tracking Case: In this section, eddy tracking is conducted through long-term, continuous observations within the specified region. The tracking performance is assessed by calculating the DE of the eddy center. Specifically, a smaller DE indicates that the detected trajectory closely approximates the true eddy trajectory. The RMSE of the eddy center DEs, calculated over the entire trajectory, is used to evaluate the overall tracking performance.

In Fig. 11, a region is selected with longitudes ranging from 151° to 154° and latitudes from 30.4° to 32° for eddy tracking, spanning from July 6 to August 10. Due to the relatively slow movement of the eddy, tracking results are presented every five days for clearer visualization of the eddy's trajectory. The eddy's motion trajectory, which is from east to west, is represented by a dashed line connecting the centers of the eddies over the past few days. From the perspective of actual

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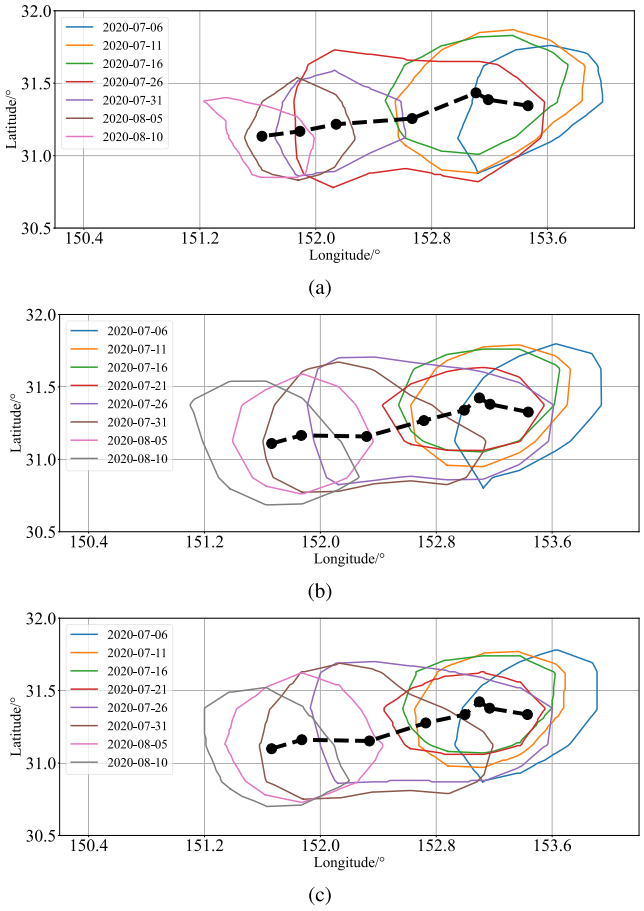


Fig. 11. (a) Tracking case using CYGNSS. (b) Same case using ACE-BOC signal. (c) Real eddy tracking result.

motion, as shown in Fig. 11(c), this eddy is classified as a large-scale eddy, with no significant changes in size observed over the course of a month.

Table II provides the eddy detection results from CYGNSS on different dates and shows the variations in different metrics through the mean and RMSE. Here, the DE was used as the evaluation metric, and the RMSE for the error of the eddy center along the path was found to be 8.40 km.

CYGNSS performance varies significantly across metrics. The SC ranges from 41.65% to 93.07%, with a mean of 65.49%, showing inconsistency. DE spans 0.67 to 20.11 km (mean: 5.84 km), indicating notable inaccuracies. AE fluctuates between 0.79 and 9.42 cm (mean: 3.88 cm), reflecting lower precision in amplitude estimation. RE ranges from 0.31 to 23.19 km (mean: 9.90 km), highlighting challenges in radius estimation. Shape error difference (ΔSE) varies from 0.48% to 29.26% (mean: 11.41%), further emphasizing inconsistency. The IRE ranges from 0.12 to 0.61, with a mean of 0.95. Overall, CYGNSS exhibits considerable variability, with significant errors in some metrics, limiting its effectiveness for applications requiring consistent precision.

Definitely, the results demonstrate a clear trajectory of movement. However, this case focuses on a relatively large eddy with an amplitude of around 50 cm, so the detection effect is more prominent. In contrast, for smaller eddies, the results may not be as reliable. Notably, on July 21, the eddy was not detected, indicating that the CYGNSS signal still requires improvement for effective eddy detection.

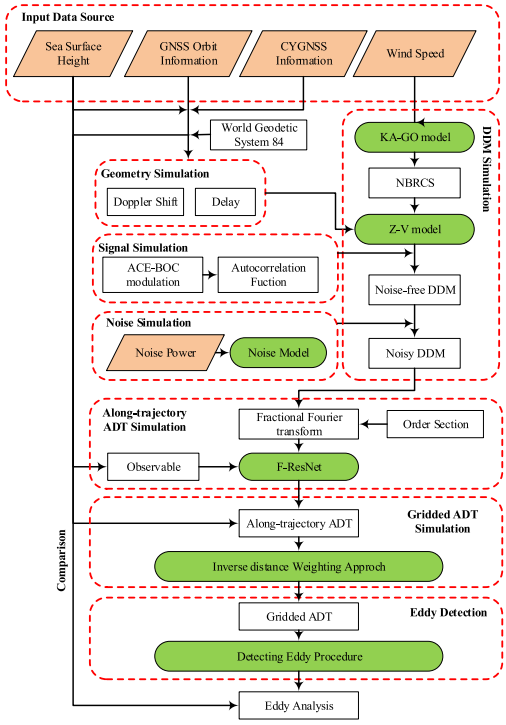


Fig. 12. Simulation flowchart for detecting ocean eddies using the ACE-BOC (15, 10) signal.

B. ACE-BOC (15, 10) Signal Simulation Data

With the modernization and improvement of GNSS, some wideband signals, which offer higher-ranging capability than the previous signals, have been designed and used [36], [37]. These signals are helpful in improving the altimetry capability of the spaceborne GNSS-R, as analyzed in [8] and [38]. At present, the widest double-sideband of GNSS signals is 50 MHz, such as the E5 AltBOC signal of the Galileo E5 and B2 ACE-BOC signal of the BeiDou. Here, the capability of detecting global eddies using the spaceborne GNSS-R with the BeiDou B2 signals is evaluated.

1) Simulation Procedure: A detailed simulation flowchart is presented in Fig. 12, illustrating the complete process for simulating the signal for eddy detection. A simulated experiment for a whole year is carried out to assess the detection of global eddies using the ACE-BOC (15, 10) signal. The GNSS orbits adopt the BeiDou IGS data, and the LEO orbits use the satellite position in the CYGNSS v3.1 data. In addition, using the CYGNSS data, the time delays caused by atmospheric and ionospheric effects were simulated based on satellite positions and respective models. Antenna information was also considered to account for time delays due to antenna baseline errors. A simulator was developed for the AEC-BOC (15, 10) signal to obtain the along-trajectory ADTs. As shown in Fig. 12, NBRCS is obtained using the KA-GO model based on wind speed data. Then, the delay and Doppler shift are estimated based on the relative positions of the

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TABLE II
COMPARED PERFORMANCE METRICS BETWEEN CYGNSS AND ACE-BOC (15, 10) OF THE TRAJECTORY

Date	CYGNSS						ACE-BOC(15,10)					
	SC/%	DE/km	AE/cm	RE/km	ΔSE/%	IRE	SC/%	DE/km	AE/cm	RE/km	ΔSE/%	IRE
Jul. 6th	85.19	3.61	3.76	0.79	29.26	0.99	93.07	1.09	0.79	2.14	0.48	0.12
Jul. 11th	70.17	2.34	4.19	9.31	1.54	0.40	91.72	0.38	0.94	2.10	0.18	0.08
Jul. 16th	75.52	1.45	3.02	6.79	0.63	0.24	91.80	0.19	0.90	1.99	0.34	0.07
Jul. 21st	-	-	-	-	-	-	97.26	0.67	0.46	0.31	0.19	0.04
Jul. 26th	86.46	6.31	1.03	3.59	0.87	0.14	95.43	1.60	0.38	1.45	0.13	0.05
Jul. 31st	41.65	20.11	1.10	23.19	1.95	0.61	88.26	1.92	0.46	3.59	0.17	0.07
Aug. 5th	55.12	1.99	4.64	12.22	28.65	0.91	84.31	0.62	0.39	3.94	2.84	0.10
Aug. 11th	44.31	5.09	9.42	13.38	16.94	3.32	85.50	1.17	1.58	4.03	8.86	0.17
Mean	65.49	5.84	3.88	9.90	11.41	0.95	90.92	0.95	0.74	2.44	1.65	0.09
RMSE	67.71	8.40	4.68	12.03	16.78	1.39	91.02	1.11	0.83	2.73	3.30	0.10

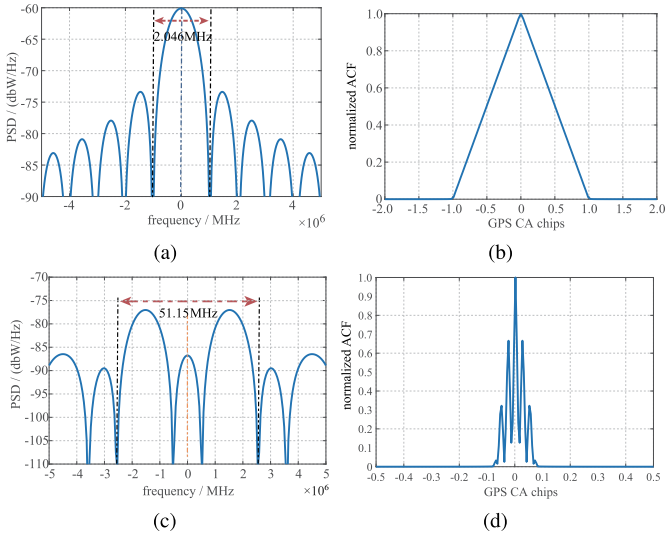


Fig. 13. (a) and (c) Power spectral densities and (b) and (d) autocorrelation functions of (a) and (b) GPS L1 CA signal and (c) and (d) ACE-BOC signal.

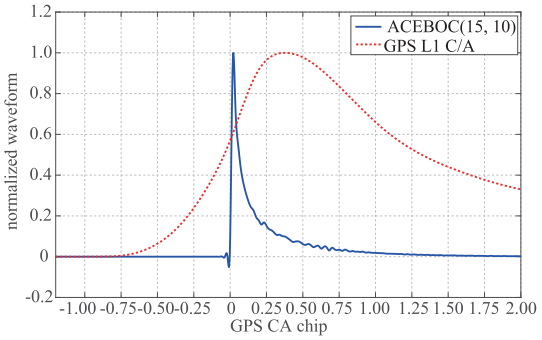


Fig. 14. Normalized delay waveforms of the GPS L1 CA and ACE-BOC (15, 10) signals for the same observation scenario.

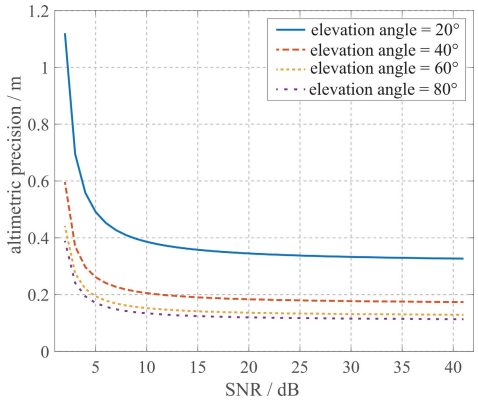


Fig. 15. Dependence of altimetric precision of the ACE-BOC (15, 10) signal on SNR and elevation angle.

receiver and transmitter. Following this, the DDM is simulated by generating an ACEBOC (15, 10) signal. Finally, noise is added according to a noise model to generate the simulated DDM [39].

The ACE-BOC (15, 10) modulation is adopted by the BeiDou B2 signal, which contains upper and lower sideband components. As shown in Fig. 13(c), two sidebands have a 30.69-MHz gap, and each sideband modulates two orthogonal signal components with a band of 20.46 MHz. The total double-sideband is 51.15 MHz, which is far wider than the 2.046-MHz double-sideband of the GPS L1 CA signal. Correspondingly, as shown in Fig. 13(d), the ACE-BOC signal has a much narrower main lobe in the autocorrelation function than the GPS L1 CA signal.

Fig. 14 shows the simulated normalized delay waveforms of the GPS L1 CA and ACR-BOC (15, 10) signals for the same observation scenario using the ZV model [40]. The delay waveform of the ACE-BOC (15, 10) signal has a further steeper lead edge than that of the GPS L1 CA signal. This steep lead edge is advantageous for tracking SSH. Fig. 15 gives the altimetric precision of the ACE-BOC (15, 10) signal, which is estimated using the theoretical model in [41]. When the elevation angle at the specular point is over 40°, and the incoherent average number is 1000, a precision better than 20 cm, which is further superior to the meter-level GPS CA scale, can obtained along the trajectory.

After the DDM is generated using the ZV model, a noise model is applied to generate a noise-free DDM. Both thermal and speckle noise are considered in the simulation, with the noise power represented as follows [42]:

$$P_n = Ti^2 kT_{\text{rec}} B_w \tag{22}$$

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TABLE III

DETECTION PERFORMANCES OF THE CYCLONIC AND ANTICYCLONIC EDDIES USING THE AEC-BOC (15, 10) SIGNAL ALTIMETRY

	cyclonic / %	anticyclonic / %
precision	85.51	84.14
Recall	91.93	91.11
F1 score	88.76	87.48
accuracy	79.80	77.75

where Ti is the coherent integration time; B_w is the signal bandwidth, determined by the coherent integration time; k is Boltzmann's constant, and T_{rec} is the receiver temperature.

After obtaining the along-trajectory ADT, the process follows the same steps as the observed CYGNSS procedure: it is input into F-ResNet, followed by gridding. Finally, the gridded global ADT is used for eddy detection and comparative analysis.

2) Result Analysis: Subsequent to the implementation of window sliding smoothing, the grid ADT demonstrates an RMSE of 0.95 cm in relation to the ground-truth values. Fig. 16 presents the detected global eddies using the ACE-BOC (15, 10) signal altimetry. The identified global eddies exhibit remarkable spatial consistency, encompassing both their position and morphology. A total of 932 802 cyclonic eddies were detected, while the actual number is 870 688. For anticyclonic eddies, the detected number is 908 404, compared with the actual number of 838 913. Of these, 800 414 cyclonic eddies and 764 311 anticyclonic eddies were correctly detected. The detection performances are given in Table III.

Fig. 17 shows the distributions of the shape errors of the ground truths and detected eddies. The similarity of the distributions indicates the spatial consistency of the identified eddies with the ground truths, and the majority of detected eddies align closely with the ground truths. This conclusion can also be derived from Fig. 18(a), which illustrates the distribution of shape error discrepancies between the ground realities and identified eddies. Most shape error differences are 5%, which are better than those of the CYGNSS-based altimetry. The shapes of the detected eddies exhibit excellent similarity, which can be clearly seen in Fig. 18(b). The SCs for both the cyclonic and anticyclonic eddies mainly range from 0.8 to 1.0, but only a few are below 0.2.

The DEs between the eddy centers of the detected eddies and ground truths are computed, and their distribution is presented in Fig. 18(c). Above 90% DEs for both the cyclonic and anticyclonic eddies are found at 0 and 20 km. Despite the grid size being 25 km, the identification of the eddy center relies on the ADT and the fit circle formed from the geostrophic flow received from the ADT. Consequently, the eddy distance inaccuracy may be less than the 25-km grid size. As presented in Fig. 18(d), the AEs between the two types of eddies mainly are within the interval of $[-2, 2]$ cm, with a positive bias of 0.4 cm, which indicates that like the CYGNSS-based altimetry, the detected eddies from the ACE-BOC (15, 10) signal code-altimetry are also stronger than the ground truths. As shown in Fig. 18(e), the REs are primarily distributed from -20 to 20 km.

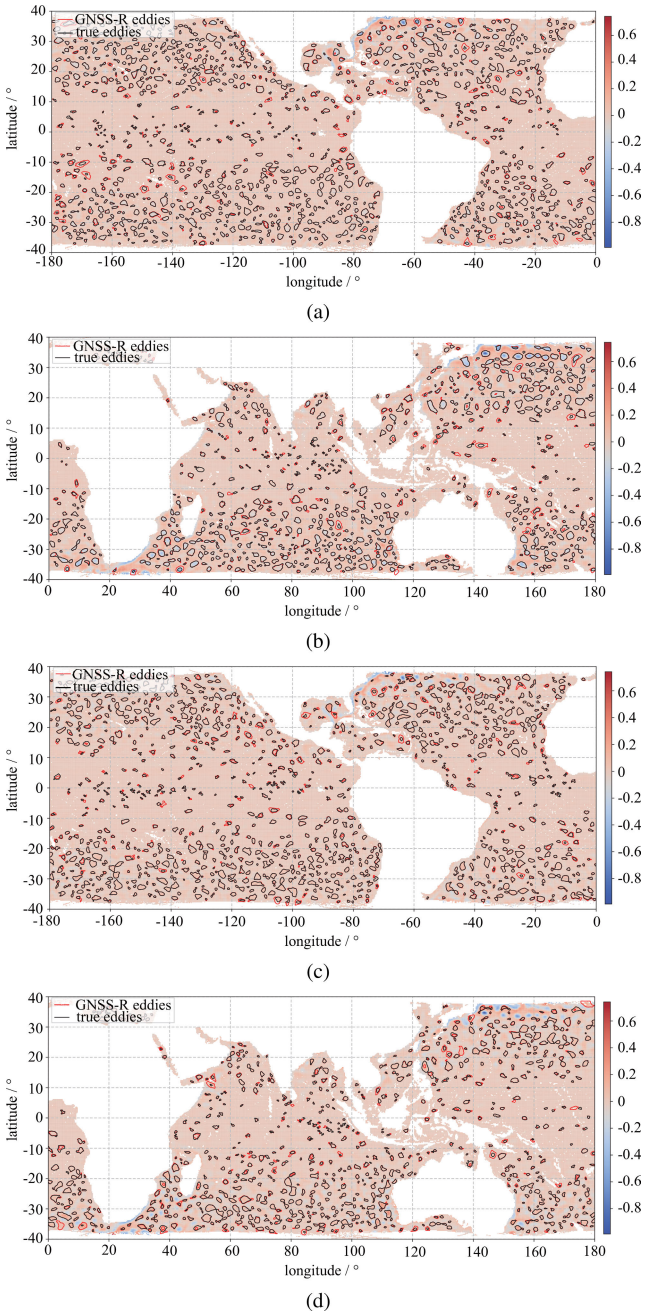


Fig. 16. Detected (a) and (b) cyclonic eddies and (c) and (d) anticyclonic eddies for (a) and (c) western hemisphere and (b) and (d) eastern hemisphere.

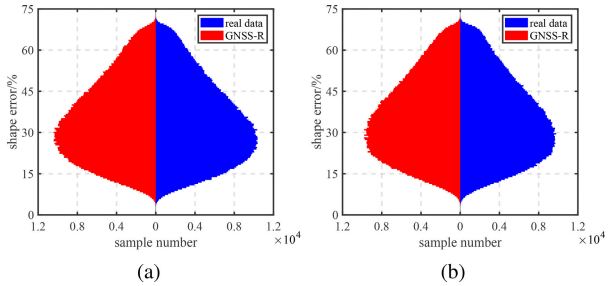


Fig. 17. Distributions of the shape errors of (a) cyclonic and (b) anticyclonic eddies.

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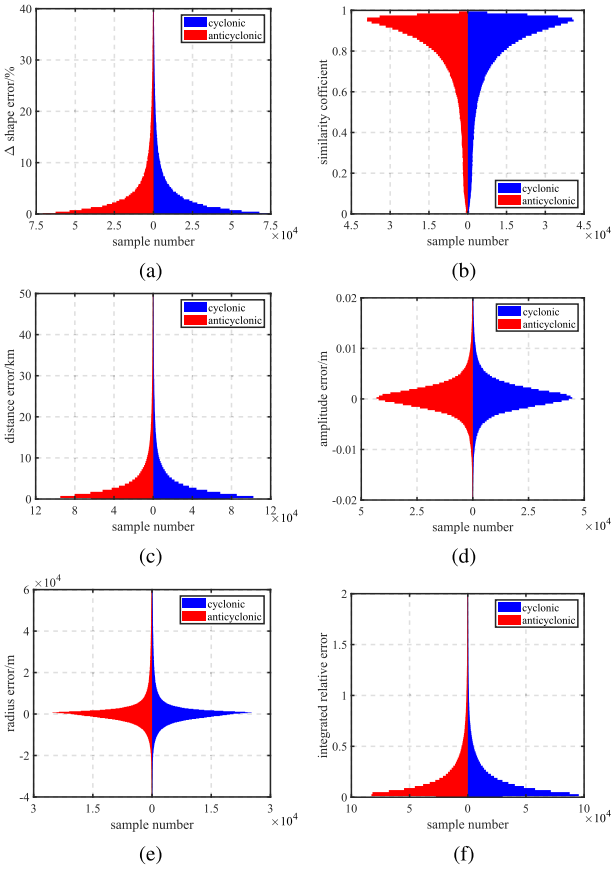


Fig. 18. Distributions of (a) shape error, (b) SCs, (c) center DEs, (d) AEs, (e) REs, and (f) IREs of the detected cyclonic and anticyclonic eddies.

Fig. 18(f) gives the statistical results of the IREs between the detected eddies and ground truths, in which most values are centered around 0.5.

3) Eddy Tracking Case: Compared with the performance of CYGNSS, the tracking performance of the same eddy using the ACE-BOC (15, 10) signal is only 1.10 km, showing a significant improvement.

ACE-BOC (15, 10) demonstrates consistently superior performance across all key metrics when compared with CYGNSS. As shown in Table II, SC for ACE-BOC (15, 10) remains high and stable, ranging from 84.31% to 97.26% with a mean of 90.92%. In terms of DE, ACE-BOC (15, 10) exhibits a narrow range from 0.38 to 1.92 km, with a mean of 0.95 km, indicating much higher precision compared with CYGNSS. AE also favors ACE-BOC (15, 10), with values between 0.39 and 1.58 cm, averaging 0.74 cm. RE for ACE-BOC (15, 10) is similarly more precise, ranging from 0.31 to 4.03 km with a mean of 2.44 km. ACE-BOC (15, 10) also outperforms CYGNSS in ΔSE, with values ranging from 0.18% to 8.86%, averaging 1.65%, while CYGNSS displays much larger variability. Furthermore, ACE-BOC (15, 10) has lower and more stable IRE, with values from 0.04 to 0.17 and a mean of 0.09, while CYGNSS shows higher variability in IRE. Overall, ACE-BOC (15, 10) offers enhanced accuracy, stability, and consistency, making it a more reliable choice for precise trajectory tracking and positioning, especially in environments where minimizing errors is critical.

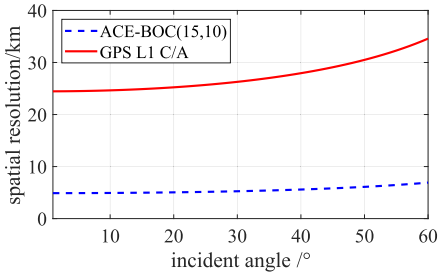


Fig. 19. Geometry of GNSS-based forward scatter radar.

V. Discussion

A. Spatial Resolution of ACE-BOC Altimeter

The ACE-BOC (15, 10) signal offers superior detection and estimation of eddies compared with the GPS L1 CA signal, making it a promising candidate for oceanic altimetry. In the aforementioned simulation, the ocean surface is gridded at 25 × 25 km^2, reflecting the spatial resolution of the GPS L1 CA signal. Spaceborne GNSS-R functions as a typical pulse-limited observation system, with spatial resolution determined by the signal bandwidth and defined as the square root of the first iso-delay ellipse [43]

$$R = \sqrt{ab} \tag{23}$$

where a and b are, respectively, the semimajor and semiminor axes of the first iso-delay ellipse. Compared with the GPS L1 CA signal, the ACE-BOC (15, 10) signal can provide a better spatial resolution. Fig. 19 gives the spatial resolutions versus the incident angle at the specular point for the GPS L1 CA and ACE-BOC signal with a height of the CYGNSS. When the incident angle is less than 30°, the spatial resolution of the GPS L1 CA signal is about 25 km. The spatial resolution of the AEC-BOC (15, 10) signal is better than 5 km. The high spatial resolution is another advantage of the ACE-BOC (15, 10) signal, which supports a denser grid than the CYGNSS-based altimetry, with 5 × 5 km^2 size, to obtain a refined eddy detection and estimation.

The real data used in this study possess a resolution of 0.25°, which corresponds precisely with the 0.25° CYGNSS data, hence ensuring uniform spatial resolution between the two. This work uses a spatial resolution of 0.25° for comparison with CYGNSS data despite the capability to get results with a resolution as fine as 0.05° under AEC-BOC (15, 10) signals, which could enhance spatial resolution and facilitate eddy detection by discerning finer characteristics. Subsequent investigations will examine analyses of greater resolution in the future.

In a pulse-limited radar system, the power originates from the first iso-delay ellipse. A diminished ellipse indicates a reduced reflected power. Consequently, under the same observational conditions and transmitted power, the reflected ACE-BOC (15, 10) signal exhibits lower power than the GPS L1 CA signal. The broader receiver bandwidth of the ACE-BOC (15, 10) signal leads to increased noise power. Fig. 15 illustrates that altimetric precision is significantly influenced by the SNR. Particularly when SNR falls below 5 dB, the

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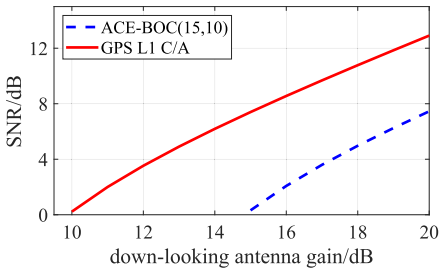


Fig. 20. SNRs of the reflected ACE-BOC (15, 10) and GPS L1 CA signals versus downlooking antenna gain.

altimetry precision deteriorates swiftly as the SNR declines. Fig. 20 illustrates the correlation between the SNR of the reflected ACE-BOC (15, 10) and GPS L1 CA signals and the gain of the downlooking antenna for the identical observational context. The SNR of the ACE-BOC (15, 10) is approximately 5 dB inferior to that of the GPS L1 CA signal. To achieve an SNR exceeding 5 dB, a downlooking antenna with a gain beyond 18 dB is required, resulting in an increase in antenna size. The reliance on the high-gain and larger downlooking antenna constitutes a limitation of the ACE-BOC (15, 10) signal altimeter.

B. Carrier-Phase Altimeter

With the development of Spire Cubesat, the capability of the spaceborne GNSS-R carrier-altimeter has been demonstrated and assessed [44]. The instruments are designed to focus on receiving reflected signals with a low incident angle, making them less effective at receiving grazing-angle signals, which are more likely to be coherent. A novel concept involving grazing-angle signals for spaceborne GNSS-R altimetry has been proposed and demonstrated [45], [46], which will benefit the precise detection of eddies.

The centimeter-level precision of the along-trajectory altimetry, which is far higher than the meter-level precision of the narrowband code-altimeter, can be obtained. This precision is beneficial to detect eddy. Due to a small number of samples of continuous phases, it is impossible to detect eddy only using the relative height from the carrier-altimeter.

The methodology derives ADT data from both satellite code-altimetry and carrier phase altimetry. The code-altimeter provides global coverage with an accuracy of approximately 12-cm along-track result, improving to a 5-cm gridded result. In contrast, the carrier-phase altimeter attains centimeter-level precision but suffers from inadequate data volume due to sea surface roughness, hindering its capacity for continuous phase tracking. To guarantee dependable outcomes, the code-altimeter is used as the foundation for global coverage, subsequently incorporating carrier phase data when accessible. Grids containing carrier phase data are prioritized for their superior accuracy, and the code-altimeter findings are substituted with the carrier phase measurements. Due to the availability of carrier phase data solely along the track, a sliding filter is used to refine the results, guaranteeing that postfiltering, the measurements attain an accuracy of within 2 cm.

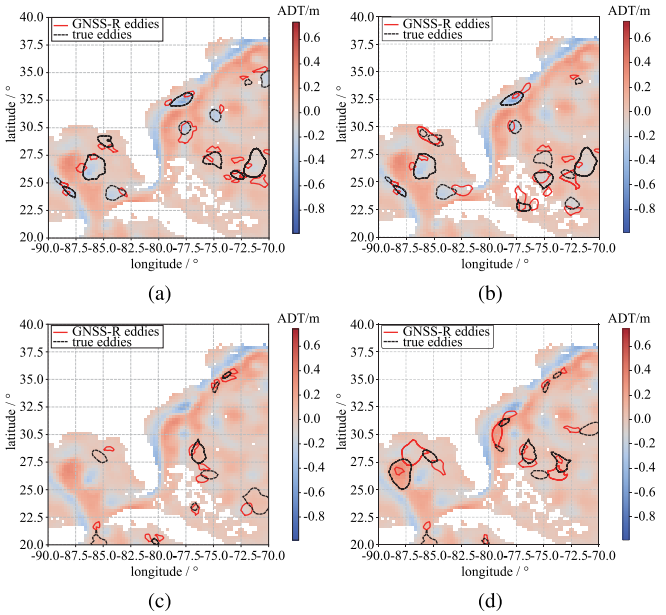


Fig. 21. Detected (a) and (b) cyclonic and (c) and (d) anticyclonic eddies using (a) and (c) only the code-altimetry and (b) and (d) two altimetry.

On each day, the CYGNSS instrument was occasionally configured to acquire the raw IF data with a sampling length of 30–60 s, which means the carrier-altimetry failed to cover the global sea surface. Here, only a regional analysis is presented, as shown in Fig. 21; for June 7, 2020, the region was longitudinally from −90° to −70° and latitudinally from 20° to 40°, respectively, where were covered by 27 raw IF data and only 10% data were effective to obtain the continuous phases. Compared with code-altimetry alone, after carrier-phase altimeter assisting, two improvements can be found. Some undetected eddies by the code-altimetry are identified, such as the anticyclonic eddies within the longitude from −77.5° to −75° and latitude from 22.5° to 25°. In addition, the eddies detected by the carrier-altimetry have better similarity with the ground truths than those from the code-altimetry. The results are presented in Table IV.

The mean SCs between the detected eddies using the carrier-altimetry assistance and the ground truths are improved to 45.21 and 48.18 from 43.81 and 40.37 for cyclonic and anticyclonic eddies, respectively. In addition, after carrier-phase altimeter assisting, the mean bias between the shape errors of the detected eddies and ground truths was also reduced. These results indicate an improvement in estimating eddy shape. The biases of the center distance, radius, and amplitude between the ground truths and the detected eddies are reduced once the carrier-altimetry is used. For example, the biases of the estimated radius decreased by 9.25% and 38.4% for the cyclonic and anticyclonic eddies, respectively, and the reductions of the amplitude are 39.68% and 47.54%, respectively. It should be noted that some detected eddies using the code-altimetry are not identified when the carrier-altimetry is applied, such as the anticyclonic eddies with the longitude from −72.5° to −70° and the latitude from 22.5° to 25°. The main reason is that the carrier-altimetry only corrects the ADT

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TABLE IV
COMPARED PERFORMANCE METRICS WITHOUT AND WITH THE CARRIER-PHASE ALTIMETER ASSISTANCE

parameters	eddy type	code-only	carrier-phase altimeter
SC/%	cyclonic	43.91	45.21
	anticyclonic	40.37	48.18
mean DE/ km	cyclonic	9.09	8.32
	anticyclonic	7.63	5.51
mean AE/ cm	cyclonic	4.36	2.52
	anticyclonic	8.12	4.26
mean RE/ km	cyclonic	9.17	7.61
	anticyclonic	7.41	5.75
mean ΔSE/ %	cyclonic	19.92	15.75
	anticyclonic	13.06	10.29
mean IRE	cyclonic	1.03	1.07
	anticyclonic	2.10	0.63

along specific trajectories rather than across the entire detected region, leading to incomplete or missing close contours.

It is important to note that while carrier-phase altimeter eddy detection improves ADT accuracy at certain locations, this may lead to a decrease in overall detection metrics for the entire oceanic detection, such as the IRE for anticyclonic eddies shown in Table IV. This can be understood as follows: if the values in certain grid cells within an eddy's contour change, the contour lines will shift, the center of the fit circle will change, and consequently, the detection metrics for the entire eddy will be affected. Therefore, the purpose of this section is to highlight improvements in certain metrics while emphasizing that specific cases should be evaluated individually.

The carrier-phase altimeter operates efficiently; however, the constraints of orbiting sensors, particularly their incapacity to estimate the carrier phase in real-time, hinder their ability to accurately evaluate global eddy detection and estimation performance.

VI. CONCLUSION

This article explored the feasibility of detecting and estimating global mesoscale eddies using spaceborne GNSS-R altimetry. An F-ResNet model was used to train the multivariate relationship between the spaceborne GNSS-R observable and the ADT, as well as the other auxiliary data for the altimetry. The CYGNSS L1 v3.1 data were used to assess the eddy detection and estimation performances of the GPS L1 CA signal. The results showed that due to its coarse ranging ability, the GPS L1 CA signal has a worse capability for eddy detection and estimation compared with radar altimetry. The precisions of the detected cyclonic and anticyclonic eddies were, respectively, only 50.63% and 47.80%, and the corresponding accuracies are, respectively, 47.92% and 45.52%. In addition, the shapes of the detected eddies had worse similarity with the ground truths. Given that the GPS L1 CA signal is inadequate to carry out the high-precision eddy detection and estimation, the ACE-BOC (15, 10) signal was taken as an example of the next-generation spaceborne GNSS-R altimeter to assess the potential of detecting and estimating eddies using the widest bandwidth GNSS signal. The simulated results showed that the ACE-BOC (15, 10) signal provided the precisions of 85.51% and 84.14% and the accuracies of 79.80% and 77.75% for the cyclonic and anticyclonic eddies, respectively. Furthermore, the shapes of the detected eddies were also in good agreement with these ground truths. Most SCs of the detected eddies with the ground truths were over 0.8. The high detection and estimation performances of the ACE-BOC (15, 10) signal suggest the necessity of using a wide-bandwidth GNSS signal to perform the next-generation spaceborne GNSS-R altimeter. This article examines the case of eddy tracking over one month, focusing on the capacity of both the signals to monitor eddies through long-term continuous observations. The findings indicate that the ACEBOC (15, 10) signal exceeds the GPS L1 signal in monitoring the eddy course. The RMSE for ACEBOC is 1.11 km, while GPS L1 has an RMSE of 8.40 km. Moreover, the GPS L1 signal fails to detect the eddy on July 21, indicating a lack of continuity. The results underscore the efficacy of the ACEBOC (15, 10) signal for accurate, continuous daily eddy detection over prolonged durations.

Finally, this article offers theoretical insights into the spatial resolution of the ACEBOC (15, 10) altimeter and the design of its antenna for practical engineering applications. A spatial resolution of 0.05° is highly valuable for future research to more accurately describe oceanic eddies. The study investigates the possibility of carrier-phase altimeter in detecting oceanic eddies. The outcomes of carrier-phase altimeter detection confirm the advancement of the grazing-angle spaceborne GNSS-R altimeter. Future research will concentrate on assessing coastal and aerial ACE-BOC signal altimeters. Furthermore, using data from ESA's PRETTY satellite, which uses carrier phase through GNSS-R signals (L1 and E5/L5) at grazing angles, may improve eddy identification precision and enrich the comprehension of small-scale marine phenomena.

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